

AI Content Transformation Platform

SIH 2026 – Comprehensive Research, Competitor Analysis & Solution Blueprint

A compact reference document for Smart India Hackathon 2026.

1. Executive Summary

Our solution is an AI-powered content transformation platform that lets operators upload any source content—PDFs, reports, articles, images, videos or prompts—and generate multiple communication formats from a single source. Unlike existing tools that generate each output independently, our platform first builds a shared **Content Spine**, ensuring every generated artefact remains accurate, consistent and faithful to the original content.

2. Understanding the SIH Problem Statement

The official SIH problem statement requires a configurable AI engine capable of transforming different types of input into multiple communication artefacts while allowing users to control audience, tone, language, detail and communication objective.

Supported Inputs

- PDFs
- Reports
- Articles
- Research Papers
- Threat Intelligence Reports
- Policy Documents
- Images
- Videos
- Free-form Prompts

Expected Outputs

- Executive Summary
- LinkedIn Post
- X Thread
- Advisory
- Presentation
- Infographic
- Complete Video Package

3. Existing Market Landscape

Category	Examples	Strength	Weakness
General AI	ChatGPT, Claude, Gemini	Powerful generation	No unified multi-output workflow
Marketing AI	Jasper, Copy.ai	Marketing content	Weak technical transformation

Presentation AI	Gamma	Great slides	Presentation only
Video AI	Synthesia	Video generation	Separate workflow
Enterprise AI	Copilot, NotebookLM	Document understanding	Limited cross-format generation

Observation: No single platform currently converts one verified source into every communication artefact while maintaining factual consistency across all outputs.

4. Competitor Analysis

ChatGPT: Versatile text generation but every output is produced independently, causing fact drift.

Claude: Excellent long-document understanding but lacks an integrated multi-output workflow.

Jasper AI: Strong marketing platform but limited for government advisories and technical reports.

Copy.ai: Fast social generation but cannot produce complete communication packages.

Gamma: Creates attractive presentations but focuses only on slides.

Synthesia: Excellent AI video creation but lacks document transformation workflows.

NotebookLM: Great research assistant but not a communication transformation engine.

Microsoft Copilot: Office integration is useful but applications still operate separately.

5. Loopholes in Existing Solutions

- Fact Drift: numbers, names and dates change across outputs.
- No Traceability: users cannot verify where information came from.
- Independent Generation: every format requires repeated prompting.
- Weak Infographics: mostly text instead of structured layouts.
- Limited Video Support: usually only scripts rather than complete production packages.

6. Our Solution

Instead of generating each output separately, our platform creates a structured Content Spine that becomes the single source of truth for every generated artefact.

Upload Once	Content Spine	Generate Outputs	Consistency Validation
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This approach eliminates repeated prompting and keeps facts synchronized across every selected output.

7. The Content Spine – Our Core Innovation

The Content Spine is a structured knowledge representation created before any output is generated. It stores entities, organizations, dates, numbers, events, risks, recommendations and relationships.

Benefits

- Preserves factual accuracy.
- Maintains consistent messaging.
- Enables simultaneous multi-output generation.
- Supports enterprise-grade reliability.

8. Advanced Features

- Fact Lock Layer that preserves dates, names and numbers.
- Explainable AI with source mapping.
- Audience Profiles for Executive, Technical, Government and Public communication.
- Complete Video Package with script, storyboard, narration, subtitles and scene descriptions.
- Consistency Validator that automatically checks generated outputs.

9. Technical Architecture

Layer	Technology
Frontend	React + TypeScript
Backend	Node.js + Express
Database	PostgreSQL
ORM	Prisma
OCR	Tesseract
Video Processing	FFmpeg
AI Models	OpenAI / Gemini / Llama
Vector Search	pgvector

10. Innovation Matrix

Feature	Existing Tools	Our Platform
Multi-output	Partial	Yes
Content Spine	No	Yes
Fact Consistency	No	Yes
Source Mapping	No	Yes
Cross-output Validation	No	Yes
Government-ready Workflow	No	Yes

11. How to Explain This to Judges

Imagine a government department receives a 50-page cyber incident report. Instead of manually creating an executive summary, advisory, LinkedIn post, presentation, infographic and awareness video, the operator uploads the report once. The platform builds a Content Spine, generates every selected output, validates factual consistency and exports ready-to-use deliverables.

Real-World Example

Input	Outputs
Cyber Threat Intelligence Report	Executive Summary, Advisory, LinkedIn, X Thread, Presentation, Infographic and Video

12. Future Scope

- Multilingual generation
- Real-time collaboration
- Approval workflows

- Compliance validation
- Organization-specific knowledge bases
- API integrations

Upload once. Generate everything. Trust every output.